

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

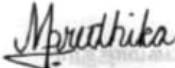
APPLICATION- BASED QUESTIONS

Code of Conduct:

I Preethika M (192511216) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:


Preethika M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

ANSWERS

Question 1

(a) Encrypted version of “MEET AT NOON” using a shift of 4

A shift-based classical technique such as the Caesar cipher encrypts a message by moving each letter forward in the alphabet by a fixed number of positions (the key). Here the key is 4, so every letter A–Z is mapped to the letter 4 positions ahead, wrapping around from Z back to A. The formula used is:

$$C = (P + 4) \bmod 26$$

where P is the numeric position of the plaintext letter (A = 0, B = 1, ... Z = 25) and C is the numeric position of the resulting ciphertext letter. Applying this rule letter-by-letter to “MEET AT NOON”:

Plaintext letter	M	E	E	T	A	T	N	O	O	N
Position (A=0)	12	4	4	19	0	19	13	14	14	13
+4 mod 26	16	8	8	23	4	23	17	18	18	17
Ciphertext letter	Q	I	I	X	E	X	R	S	S	R

Reassembling the ciphertext letters in the original word structure (4 letters, space, 2 letters, space, 4 letters) gives the encrypted message:

MEET AT NOON → QIIX EX RSSR

(b) How an authorized member retrieves the original message

An authorized member who knows the encoding rule (a shift of 4) simply reverses the process by shifting each ciphertext letter backward by 4 positions instead of forward. This is equivalent to applying the inverse formula:

$$P = (C - 4 + 26) \bmod 26$$

The addition of 26 before taking the modulus prevents negative values when C is smaller than 4 (for example, decrypting the letter ‘E’). Working through the ciphertext “QIIX EX RSSR”: Q(16)–4=12=M, I(8)–4=4=E, I(8)–4=4=E, X(23)–4=19=T, E(4)–4=0=A, X(23)–4=19=T, R(17)–4=13=N, S(18)–4=14=O, S(18)–4=14=O, R(17)–4=13=N. This exactly recovers the original message “MEET AT NOON”. Since both parties share only the single numeric key (4), the scheme relies entirely on the secrecy of that shift value — which is also its major weakness, because the key space contains only 25 possible shifts and can be brute-forced almost instantly.

Question 2

(a) *Deducing the most likely original message*

The intercepted text “GSRH RH Z HVXIVG” shows the structural signature of the Atbash cipher, a classical monoalphabetic substitution in which each letter is replaced by its mirror-image letter in the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, and so on). The governing relationship is:

$$\text{Cipher position} = 25 - \text{Plain position}$$

Ciphertext letter	G	S	R	H	R	H	Z	H	V	X	I	V	G
Position (0–25)	6	18	17	7	17	7	25	7	21	23	8	21	6
25 – position	19	7	8	18	8	18	0	18	4	2	17	4	19
Plaintext letter	T	H	I	S	I	S	A	S	E	C	R	E	T

Decoding every letter in this way recovers the plaintext:

GSRH RH Z HVXIVG → THIS IS A SECRET

(b) *Justification using pattern analysis*

Several observations point conclusively to the Atbash cipher rather than a random or complex substitution:

1. Symmetry check: For every letter pair (cipher, plain) found above, the two numeric positions add up to exactly 25 (e.g., $G=6$ and $T=19$, $6+19=25$; $Z=25$ and $A=0$, $25+0=25$). This constant-sum property is the mathematical signature unique to Atbash and would not hold for an arbitrary Caesar shift or a keyed substitution.

2. Word-length pattern: The ciphertext breaks into groups of 4, 2, 1 and 6 letters, matching common short English words. Once decoded this becomes “THIS / IS / A / SECRET”, a natural sentence pattern with a single-letter word (‘A’) which is a strong linguistic cue in English cryptanalysis.

3. Repeated letter behaviour: The letter ‘H’ appears three times in the ciphertext and always decodes to the same plaintext letter ‘S’, confirming a consistent one-to-one monoalphabetic mapping rather than a polyalphabetic scheme.

4. Double-letter structure: The ciphertext contains no doubled letters, but the plaintext “SECRET” contains a repeated ‘E’ at non-adjacent positions — consistent with normal English letter frequency once decoded, reinforcing that the decoding is linguistically valid.

Together, these observations confirm both the cipher type (Atbash) and the correctness of the recovered plaintext “THIS IS A SECRET.”

Question 3

(a) Step-by-step Vigenère encryption of “HELLO” with keyword “KEY”

The Vigenère cipher is a polyalphabetic technique that encrypts each plaintext letter using a different Caesar shift, determined by the corresponding letter of a repeating keyword. Since the plaintext “HELLO” has 5 letters and the keyword “KEY” has only 3, the keyword is first repeated cyclically until it matches the plaintext length:

Plaintext : H E L L O
Keyword : K E Y K E

Each letter pair is then combined using the formula:

$$C = (P + K) \bmod 26$$

Plaintext letter (P)	H	E	L	L	O
Keyword letter (K)	K	E	Y	K	E
P value (A=0)	7	4	11	11	14
K value (A=0)	10	4	24	10	4
(P+K) mod 26	17	8	9 (35-26)	21	18
Ciphertext letter	R	I	J	V	S

The resulting ciphertext is:

HELLO → RIJVS

(b) How the repeating keyword influences each character

Because the keyword “KEY” is shorter than the plaintext, it is repeated cyclically (K-E-Y-K-E-...) so that every plaintext letter is paired with exactly one keyword letter. Each keyword letter defines a distinct Caesar shift for the position it occupies: ‘K’ (value 10) shifts a letter forward by 10, ‘E’ (value 4) shifts by 4, and ‘Y’ (value 24) shifts by 24. Consequently, the two occurrences of the letter ‘L’ in “HELLO” are encrypted differently — the first ‘L’ is paired with ‘Y’ and becomes ‘J’, while the second ‘L’ is paired with ‘K’ and becomes ‘V’. This is the key strength of the Vigenère cipher over simple substitution: identical plaintext letters do not necessarily map to identical ciphertext letters, which flattens the letter-frequency distribution and defeats basic frequency analysis. However, because the keyword repeats periodically, the cipher remains vulnerable to attacks such as the Kasiski examination or index-of-coincidence analysis, which can recover the keyword length and subsequently break the cipher when enough ciphertext is available.

Question 4

(a) How investigators detect hidden data within the logo image

Steganography conceals data inside a cover file so that its presence is not obvious on casual viewing, but the act of embedding almost always leaves detectable statistical or structural traces. Digital forensic investigators typically use the following analytical approaches, collectively known as steganalysis:

1. Visual and structural inspection: Comparing the suspect logo file against a known-clean original copy (if available) to spot subtle differences in colour banding, unusual noise, or irregular pixel patterns invisible to the naked eye but revealed by contrast stretching or bit-plane slicing.
2. Statistical analysis: Techniques such as the Chi-square attack examine whether the frequency distribution of pixel values matches what is expected from an unmodified image. Least-Significant-Bit (LSB) embedding tends to equalise pairs of values (Pairs-of-Values analysis), producing statistically detectable anomalies.
3. Histogram analysis: Plotting the colour/intensity histogram of the image; genuine photographs show smooth, continuous distributions, whereas LSB-modified images often show unnatural spikes or flattening at certain intensity levels.
4. File size and metadata anomalies: An image containing hidden data may be unusually large for its visual complexity, or may show inconsistencies in metadata, header structure, or compression artefacts compared to a standard file of the same format.
5. Automated steganalysis tools: Tools such as StegExpose, StegDetect, or machine-learning-based classifiers analyse multiple statistical features simultaneously and flag the probability that a file contains hidden payloads.

(b) Two techniques to strengthen hidden communication against detection or extraction

1. Encrypt the payload before embedding (hybrid crypto-steganography): Rather than hiding the raw confidential data directly, the sender should first encrypt it using a strong symmetric algorithm such as AES, and only then embed the resulting ciphertext into the cover image. This provides defence-in-depth: even if steganalysis successfully detects and extracts the hidden bit-stream, the extracted data remains unreadable without the encryption key. This combination of confidentiality (encryption) and covertness (steganography) is significantly stronger than either technique used alone.
2. Use transform-domain and adaptive embedding instead of simple spatial-domain LSB: Embedding data directly into the least significant bits of raw pixels (spatial domain) is the easiest form of steganography to detect statistically. Security can be substantially improved by embedding in the transform domain — for example, modifying quantised DCT coefficients (as in JPEG steganography) or DWT coefficients — combined with adaptive/randomised pixel selection using a pseudo-random number generator seeded by a shared secret key. This spreads the hidden bits unpredictably across the image (rather than sequentially) and concentrates changes in perceptually less-sensitive regions, which greatly reduces the statistical footprint that steganalysis tools rely on.

Question 5

(a) Reversing the transposition to find the original message

The phrase “three levels of arrangement” indicates a Rail Fence (zig-zag) transposition cipher written across 3 rails. In this scheme the plaintext is written diagonally in a zig-zag pattern across 3 rows, and the ciphertext is produced by reading the rows one after another (row 0, then row 1, then row 2). Since the ciphertext “WECRLTEERDSOEFEAOCAIVDEN” has 25 characters, the row lengths for a 3-rail fence follow a repeating cycle of 4 positions per column-pair, giving 7 characters in row 0, 12 in row 1, and 6 in row 2 ($7+12+6 = 25$). Splitting the ciphertext accordingly:

```
Row 0 (7 letters):  W E C R L T E
Row 1 (12 letters): E R D S O E E F E A O C
Row 2 (6 letters):  A I V D E N
```

These three rows are now placed back into the zig-zag diagonal pattern (down-up-down across rows 0-1-2-1-0-1-2...) to rebuild the original diagonal writing order:

```
Pos:  0  1  2  3  4  5  6  7  8  9 10 11 12
Row0: W  .  .  .  E  .  .  .  C  .  .  .  R
Row1: .  E  .  R  .  D  .  S  .  O  .  E  .
Row2: .  .  A  .  .  .  I  .  .  .  V  .  .
```

```
Pos: 13 14 15 16 17 18 19 20 21 22 23 24
Row0: .  .  .  L  .  .  .  T  .  .  .  E
Row1: E  .  F  .  E  .  A  .  O  .  C  .
Row2: .  D  .  .  .  E  .  .  .  N  .  .
```

Reading each column from top to bottom, position by position (0 to 24), reconstructs the plaintext:

W-E-A-R-E-D-I-S-C-O-V-E-R-E-D-F-L-E-E-A-T-O-N-C-E

which, when broken into words, gives the original message:

“WE ARE DISCOVERED. FLEE AT ONCE.”

(b) Why rail fence / transposition is vulnerable in modern systems

1. Extremely small key space: The only “key” is the number of rails. For a message of reasonable length, an attacker only needs to try a handful of small integers (2, 3, 4, 5 ... rails) and check which produces readable text — an exhaustive search takes seconds even by hand, and microseconds by computer.
2. No confusion, only diffusion of position: A transposition cipher merely rearranges the existing plaintext letters; it never substitutes or alters their identity. This means the ciphertext always contains exactly the same letter-frequency distribution as the plaintext, so classical frequency analysis instantly confirms the underlying language and can guide reconstruction.
3. Vulnerable to anagramming and pattern recognition: Because letters are simply reordered, tools that test common word fragments, digrams, and trigrams (e.g., ‘TH’, ‘ING’, ‘THE’)

against candidate rail counts can reconstruct the message very quickly, especially when the plaintext is natural-language prose rather than random data.

4. No resistance to known-plaintext or brute-force attacks: If an attacker has even a partial guess of the message content (a crib), the exact rail arrangement can be deduced almost immediately, which is unacceptable for modern confidentiality requirements where adversaries routinely have partial knowledge of communications.

(c) Proposed enhancement to improve confidentiality

The most effective and practical improvement is to convert the cipher into a product cipher by combining transposition with substitution — the same principle modern block ciphers (like AES) are built on. Concretely, the organisation should first encrypt the message using a substitution-based algorithm (for example, a keyed Vigenère cipher or, ideally, a modern symmetric cipher such as AES), and only then apply the rail-fence/columnar transposition to the resulting ciphertext (or vice-versa), repeating the substitution-transposition cycle for multiple rounds. This hybrid approach ensures that:

- Substitution destroys the direct letter-frequency correspondence between plaintext and ciphertext (confusion).
- Transposition scatters the substituted letters so that positional and word-pattern clues are removed (diffusion).
- The combined key space (substitution key $\times$ number of transposition arrangements) is vastly larger than either technique alone, making brute-force and frequency-based attacks impractical.

In practice, legacy systems should be migrated to a standard modern symmetric algorithm (AES-256 in GCM mode) for actual message confidentiality and integrity, using the classical product-cipher principle only as a conceptual bridge to explain why layering substitution and transposition strengthens legacy infrastructure until full migration is complete.